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Inventor(s)

Schlack; Richard E.

MODULAR LATCH SYSTEM

Abstract

A compression latch for a door includes a driver that is rotatable with respect to the door between an unlocked position and a locked position. The driver is configured to be mounted to one side of the door. A driver shaft is non-rotatably connected to the driver such that the driver shaft rotates along with the driver. In an unlocked position of the driver, the pawl is positioned to permit opening of the door, and, in a locked position of the driver, the pawl is positioned to prevent opening of the door. The compression latch is configured to translate the pawl closer toward the door upon rotating the driver from the unlocked position to the locked position.

Inventors: Schlack; Richard E. (Concordville, PA)

Applicant: Southco, Inc. (Concordville, PA)

Family ID: 74595469

Assignee: Southco, Inc. (Concordville, PA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 17/794,382, filed Jul. 21, 2022, which is the U.S. National Phase Patent Application of PCT Patent Application No. PCT/US2021/014985, filed Jan. 26, 2021, which claims priority to U.S. Provisional Patent Application No. 62/968,464, filed on Jan. 31, 2020, and U.S. Provisional Patent Application No. 63/084,205, both titled “MODULAR LATCH SYSTEM,” the entireties of all of which is incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present invention relates to the field of mechanical latches.

BACKGROUND OF THE INVENTION

[0003] As is described in U.S. Pat. No. 7,441,812 to Southco, Inc., compression latches for mounting on doors or panels are known. Compression latches are used in applications in which it is desirable to both latch a door or panel to the frame in which it is mounted and to seal the edge of the panel to the frame when closed. For example, compression latches are desirable when the opening in which the panel is mounted is provided with a gasket that must be compressed to provide a seal. Compression latches may be used on doors for heating, ventilation and air conditioning (HVAC) units, for example. HVAC unit doors, which are typically thick and filled with insulation, can have wide thickness tolerances. It would be desirable to provide a modular compression latch that compensates for the thickness tolerance.

SUMMARY OF THE INVENTION

[0004] According to a first aspect of the present invention, a compression latch for a door is provided. The latch includes a driver that is rotatable with respect to the door between an unlocked position and a locked position. The driver is configured to be mounted to one side of the door. A driver shaft is non-rotatably connected to the driver such that the driver shaft rotates along with the driver. A pawl assembly comprises (i) a housing that is configured to be fixedly mounted to the door, and (ii) a pawl that is movably connected to the housing and is configured to both rotate and translate with respect to the housing in response to rotation of the driver shaft. In an unlocked position of the driver, the pawl is positioned to permit opening of the door, and, in a locked position of the driver, the pawl is positioned to prevent opening of the door. The compression latch is configured to translate the pawl closer toward the door upon rotating the driver from the unlocked position to the locked position.

[0005] According to another aspect of the present invention, a pawl assembly for locking a door is provided. The pawl assembly comprises a housing that is configured to be fixedly mounted to the door, and a pawl that is movably connected to the housing and is configured to both rotate and translate with respect to the housing. In an unlocked position of the pawl assembly, the pawl is positioned to permit opening of the door, and, in a locked position of the driver, the pawl is positioned to prevent opening of the door. The pawl is configured to move closer to the door upon moving the pawl assembly from the unlocked position to the locked position.

[0006] According to still another aspect of the present invention, a driver is configured to be rotatably mounted to a door. The driver defines a handle that is capable of being grasped by a user.

An opening is formed in the driver. A fastener is positioned within the opening for mounting the driver to the door. A cover is removably mounted to the driver and is positioned over the opening to conceal a head of the fastener, the cover defining a relief in the form of an undercut on a surface thereof. A lock plug is rotatably mounted to the driver between an unlocked state and a locked state, wherein in the unlocked state of the lock plug, the driver can be rotated with respect to the door, and in the locked state of the lock plug, the driver is prevented from rotating with respect to the door. A projection extends from the lock plug, the projection being non-rotatable with respect to the lock plug. In a locked state of the lock plug, the projection is positioned within the relief to prevent removal of the cover from the driver and consequent access to the fastener, and in an unlocked state of the lock plug, the projection is not positioned within the relief to permit removal of the cover from the driver and allow consequent access to the fastener.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The above and other aspects and features of the present invention will become more apparent to those of ordinary skill in the art by describing in detail exemplary embodiments thereof with reference to the attached drawings.

[0008] FIGS. 1A-1C depict isometric, top plan and cross-sectional views, respectively, of a compression latch assembly (latch, hereinafter) that is mounted to the door, wherein the latch is shown rotated to an unlocked state.

[0009] FIGS. 2A-2C depict isometric, top plan and cross-sectional views, respectively, of the door mounted latch of FIG. 1A, wherein the latch is shown rotated to a partially locked state.

[0010] FIGS. 3A-3C depict isometric, top plan and cross-sectional views, respectively, of the door mounted latch of FIG. 1A, wherein the latch is shown rotated to a locked and compressed state.

[0011] FIG. 4 is an exploded view of the latch of FIG. 1A.

[0012] FIG. 5 is an exploded view of a pawl sub-assembly of the latch of FIG. 1A.

[0013] FIGS. 6A-6H are isometric, rear elevation, right side elevation, front elevation, left side elevation, bottom plan, top plan and enlarged cross-sectional views, respectively, of a driver of the latch of FIG. 1A.

[0014] FIGS. 7A-7E are isometric, bottom plan, side elevation, top plan and front elevation views, respectively, of a housing of the pawl sub-assembly of FIG. 5.

[0015] FIGS. 8A-8D are isometric, side elevation, bottom plan, and cross-sectional views, respectively, of a sleeve of the pawl sub-assembly of FIG. 5.

[0016] FIGS. 9A-9E are isometric, front elevation, side elevation, top plan, and cross-sectional views, respectively, of a threaded shaft of the pawl sub-assembly of FIG. 5.

[0017] FIGS. 10A-10E are isometric, right elevation, left elevation, front elevation, and rear elevation views, respectively, of a driver shaft of the latch of FIG. 1A.

[0018] FIGS. 11A-11E are isometric, front elevation, side elevation, rear elevation, and cross-sectional views, respectively, of a bezel of the latch of FIG. 1A.

[0019] FIGS. 12A and 12B depict isometric views of a multi-point latch system for securing a door.

[0020] FIGS. 13A and 13B depict exploded and assembled views of a top portion of a linkage of the multi-point latch system of FIG. 12A.

[0021] FIG. 14A is an exploded view of the driver of the latch of FIG. 1A.

[0022] FIGS. 14B-14F show views of the driver of the latch of FIG. 1A in an unlocked state, and FIGS. 14G-14K show views of the driver in a locked state.

[0023] More particularly, FIGS. 14B and 14C are partial side-elevation views of the driver of FIG. 14A shown in an unlocked state. FIGS. 14D and 14E are cross-sectional views of the driver of FIGS. 14B and 14C taken along the lines 14D-14D and 14E-14E, respectively. FIG. 14F is a

detailed view of FIG. 14E.

[0024] FIGS. 14G and 14H are partial side-elevation views of the driver of FIG. 14A shown in a locked state. FIGS. 14I and 14J are cross-sectional views of the driver of FIGS. 14G and 14H taken along the lines 14I-14I and 14J-14J, respectively. FIG. 14K is a detailed view of FIG. 14J.

DETAILED DESCRIPTION OF THE INVENTION

[0025] Although the invention is illustrated and described herein with reference to specific embodiments, the invention is not intended to be limited to the details shown. Rather, various modifications may be made in the details within the scope and range of equivalents of the claims and without departing from the invention.

[0026] Various terms are used throughout the disclosure to describe the physical shape or arrangement of features. A number of these terms are used to describe features that conform to a cylindrical or generally cylindrical geometry characterized by a radius and a center axis perpendicular to the radius. Unless a different meaning is specified, the terms are given the following meanings. The terms “longitudinal”, “longitudinally”, “axial” and “axially” refer to a direction, dimension or orientation that is parallel to a center axis. The terms “radial” and “radially” refer to a direction, dimension or orientation that is perpendicular to the center axis. The terms “inward” and “inwardly” refer to a direction, dimension or orientation that extends in a radial direction toward the center axis. The terms “outward” and “outwardly” refer to a direction, dimension or orientation that extends in a radial direction away from the center axis.

[0027] In the description, relative terms such as “horizontal,” “vertical,” “up,” “down,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and normally are not intended to require a particular orientation.

[0028] Terms concerning attachments, coupling and the like, such as “mounted,” “connected” and “interconnected,” refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise.

[0029] The terms “proximal” and “distal” are herein used throughout this disclosure as relative terms. In the context of describing the latch, the term “proximal,” such as in the phrase “proximal end”, is herein intended to mean toward or closer to a driver 14 of the latch, whereas the term “distal,” such as in the phrase “distal end” is herein intended to mean away from or further away from driver 14 of the latch.

[0030] FIGS. 1A-4 depict a compression latch assembly 10 (latch 10, hereinafter) that is mounted to a door 12. FIGS. 1A-1C depict the latch 10 in an unlocked state, FIGS. 2A-2C depict the latch 10 in a partially locked state, and FIGS. 3A-3C depict the latch 10 in a locked and compressed state.

[0031] Latch 10 generally comprises a driver sub-assembly 11 that is connected to a pawl sub-assembly 33 for maintaining a door 12 in either locked or unlocked state with respect to a frame to which door 12 is movably mounted. Door 12 includes a front side 12a, a rear side 12b, and a hollow area positioned between those sides, which may be filled with foam, for example. Door 12 may vary from that which is shown and described.

[0032] Referring now the features of driver sub-assembly 11, driver sub-assembly 11 generally comprises a driver 14 that is (i) rotatably mounted to a cylindrical bezel 16 about longitudinal axis A, and (ii) non-rotatably connected to a driver shaft 20.

[0033] Driver 14 is shown in FIGS. 6A-6H. Driver 14 is an L-shaped member that includes an elongated portion having a gripping surface 13 for grasping by an end user. Driver 14 is rotatably mounted to bezel 16 about longitudinal axis A, however, it should be understood that driver 14 is prevented from translating along axis. Driver 14 may be provided in the form of a handle, a knob, a lever, a ‘T’, or a tool driver, for example.

[0034] A lock plug **40** is mounted within an opening on the front face of driver **14**. Lock plug **40**, which has a recess for receiving a key or other tool, is configured to selectively permit rotation of driver **14**, as is known in the art. Lock plug **40** is an optional component of driver **14**.

[0035] In a locked configuration of lock plug **40**, rotation of driver **14** is not possible. And, in an unlocked configuration of lock plug **40**, rotation of driver **14** is possible. More particularly, as shown in FIGS. **6H** and **4**, when the proper key is inserted into the lock plug **40**, the user can rotate the lock plug **40**. A cam or ramp **42** on the underside of the lock plug **40** rotates therewith and interacts with a post **44** that is loaded by a spring **46**. Post **44** is translatable positioned with a recess **48** formed in the underside of driver **14**.

[0036] In the locked position of lock plug **40** (shown in FIG. **6H**), a bearing surface of ramp **42** bears on the proximal end **44a** of post **44** and moves post **44** in the direction shown by the arrow in FIG. **6H** against the bias of spring **46**. And, the distal end **44b** of post **44** is positioned within opening **27** (FIG. **4**) of stationary bezel **16**. When distal end **44b** of post **44** is positioned within opening **27** of stationary bezel **16** it is not possible to rotate driver with respect to bezel **16**.

[0037] In the unlocked position of lock plug **40** (not shown), an opening in ramp **42** is rotationally aligned with post **44** such that ramp **42** does not bear on the proximal end **44a** of post **44**. Thus, spring **46** is permitted to move post **44** in an upward direction such that distal end **44b** of post **44** does not protrude from the lower side of driver **14**. Accordingly, post **44** is not positioned within opening **27** of stationary bezel **16**. When distal end **44b** of post **44** is not positioned within opening **27** of stationary bezel **16** it is possible to rotate driver with respect to bezel **16** (as well as door **12**).

[0038] A cover **54** is mounted to the front face of driver **14** to conceal the head of fastener **34** in an effort to prevent unauthorized removal of the fastener **34**. Cover **54** includes an opening **56** to provide access to the front face of lock plug **40**. As will be described with reference to FIGS. **14A-14K**, the lock plug **40** and cover **54** are specially configured to prevent removal of the cover **54**, as well as the underlying fastener **34**, in a locked state of the lock plug **40**. In an unlocked state of the lock plug **40**, the cover **54**, as well as the underlying fastener **34**, may be removed/disassembled from the driver **14**.

[0039] Turning now to FIGS. **14A-14K**, the cover **54** includes a protrusion **55** on its interior facing surface that is positioned to extend through the opening **36** in the driver **14**. A relief **57** is formed in the bottom side of the protrusion **55**. The relief **57** may also be referred to herein as a cutout, gap, slot, opening, surface or recess.

[0040] A cam lock **41** is non-rotatably mounted to the body of the lock plug **40** such that the cam lock **41** rotates along with the lock plug **40**. The cam lock **41** is situated behind a shoulder formed on the front face of the lock plug **40** such that, in an assembled configuration of the driver **14**, the cam lock **41** is not visible to an end user. The cam-lock **41** is (optionally) a metallic member having a uniform thickness (as viewed in FIG. **14E**), a tear drop shaped perimeter (more generally, non-circular perimeter), and an opening formed in the center of the tear drop shape through which the lock plug **40** is mounted. The rounded perimeter portion of the tear drop shaped perimeter extends further from the axis of rotation of the cam lock **41** than does the flat perimeter portion **41a** of the tear drop shaped perimeter.

[0041] In operation of the lock plug **40**, the cam lock **41** is positioned to selectively enter the relief **57** of the cover **54**. More particularly, in a locked state of the lock plug **40** shown in FIGS. **14G-14K**, the lock plug **40** is rotated to a position such that the cam lock **41** is positioned within the relief **57**. In particular, in the locked state, the rounded perimeter portion of the tear drop shaped perimeter is positioned within the relief **57**. And, in an unlocked state of the lock plug **40** shown in FIGS. **14B-14F**, the lock plug **40** is rotated to a position such that the cam lock **41** is not positioned within the relief **57**. In particular, in the unlocked state, the flat perimeter portion **41a** of the tear drop shaped perimeter is rotationally aligned with but radially spaced apart and separated from the relief **57**.

[0042] In the locked state of the lock plug **40**, it is not possible to remove the cover **54** (and access

the fastener **34**) because the cover **54** is captivated to the driver **14** by the cam lock **41**. And, in the unlocked state of the lock plug **40**, it is possible to remove the cover **54** (and access the fastener **34**) because the cover **54** is not captivated to the driver **14** by the cam lock **41**.

[0043] Those skilled in the art will recognize that the cam lock **41** is not limited to that which is shown in the figures. For example, the cam lock **41** may be a protrusion that projects radially from the lock plug **40** for selectively interacting with the relief **57**. The cam lock **41** may be referred to herein as a projection. The projection may be either integral with or separate from the lock plug **40**.

[0044] Bezel **16** is shown in FIGS. **11A-11E**. Bezel **16** is fixedly mounted to a front side **12a** of door **12** by two self-tapping fasteners **18** that are capable of piercing door **12**.

[0045] Each fastener **18** is positioned through a respective hole **17** formed in bezel **16**. It should be understood that fasteners **18** as well as any holes for receiving fasteners **18** can vary, as is known in the art. An alignment projection **25** (FIGS. **11E** and **11D**) on the underside of bezel **16** engages with a hole **15** in door **12** to key the rotational position of bezel **16** on door **12** and also prevent rotation of bezel **16** as driver **14** moves between the locked and unlocked states. Opening **27** is formed on the front face of bezel **16** and is configured to interact with a driver locking feature, as was described above.

[0046] Driver shaft **20** is rotatably mounted to bezel **16** and is non-rotatably mounted to driver **14**. Driver shaft **20** is also shown in FIGS. **10A-10E**. The term “non-rotatable” means, for example, that driver shaft **20** is not rotatable with respect to driver **14**, however, driver **14** and driver shaft **20** can simultaneously rotate in the same direction. A non-circular head portion **22** (i.e., having opposing flat portions **31**) is provided at the proximal end of the driver shaft **20**. Head portion **22** is non-rotatably positioned within a non-circular recess **24** (FIG. **6B**) formed in the rear side of driver **14**. A cylindrical shaft portion **26** of driver shaft **20** is rotatably positioned within a central hole **23** (FIG. **11B**) of bezel **16** such that driver shaft **20** can rotate within bezel **16**. The non-circular connector **28** at the distal end of driver shaft **20** is configured to pass through an opening **30** in door **12** and engage with the pawl subassembly **33**, as will be explained in greater detail with respect to pawl subassembly **33**. The non-circular connector **28** of driver shaft **20** is cylindrical and has two flat portions **29**, as viewed in cross-section, for keying driver shaft **20** to pawl subassembly **33**. A threaded fastener **34** is positioned through an opening **36** in the front face of driver **14** and is configured to be mounted to a threaded opening **38** formed within head portion **22** of driver shaft **20**. Fastener **34** fixedly mounts driver **14** to shaft **20**; and, driver **14** and driver shaft **20** are held non-rotatable due to the keyed interface between non-circular head portion **22** and non-circular recess **24**.

[0047] Referring now to pawl sub-assembly **33** of latch **10**, which is shown on the right hand side of door **12** in FIG. **4** and in FIG. **5**, pawl sub-assembly **33** includes a hollow cylindrical housing **60** that is fixedly and non-rotatably mounted to the rear side **12b** of door **12** by fasteners **62**. Housing **60** is shown in FIGS. **7A-7E**. Housing **60** remains stationary during operation of latch **10**. Specifically, housing **60** has shoulders or flanges **64** at its proximal end (i.e., door mounting end), and the shoulders or flanges **64** include holes **61** through which the fasteners **62** are positioned. Like fasteners **18**, fasteners **62** may be self-tapping screws that are capable of piercing holes in door **12**. Fasteners **62** can vary. A central hole **65** oriented with axis A is defined in the proximal end. In an assembled form of latch **10**, connector **28** of driver shaft **20** is positioned through hole **65** of housing **60**.

[0048] Two opposing ramps **66a** and **66b** (referred to either individually or collectively as ramp(s) **66**) are positioned on opposite sides of the revolved side wall of housing **60**. Ramps **66** are symmetrical (and mirror images) and spaced apart in a circumferential direction by 180 degrees. Each ramp **66** extends in a circumferential direction about the axis A. Each ramp **66** includes a proximal portion **67a**, a distal portion **67b**, and an intermediate portion **67c** defined between portions **67a** and **67b**. As viewed in an axial direction, proximal portion **67a** extends in the proximal direction (as compared to portions **67b** and **67c**). Portion **67c** is a detent that is provided

for pressure-reduction purposes, as will be described in greater detail later.

[0049] Each ramp **66** may pass through either all or a portion of the wall thickness of housing **60**. Alternatively, each ramp **66** could be provided in the form of a projection. Ramps **66** may also be referred to as slots, guides or cams. Ramps **66** are sized to accommodate pins **70a** and **70b** (referred to either individually or collectively as pin(s) **70**) in a form-fitting manner, as will be described in greater detail later.

[0050] A sleeve **72** in the form of a hollow cylindrical body is positioned within the hollow interior of housing **60**. Sleeve **72** is depicted in FIGS. **8A-8D**. A non-circular hole **74** is defined on the proximal end face of sleeve **72**. Hole **74** is sized to accommodate connector **28** of driver shaft **20**. Hole **74** and connector **28** are keyed to each other such that driver shaft **20** is non-rotatable with respect to sleeve **72**. In other words, sleeve **72** rotates with driver shaft **20**. A large diameter circular hole **76** is defined on the distal end face of sleeve **72**. A circular head **82** of threaded shaft **80** is configured to be positioned within hole **76**. Holes **74** and **76** are co-aligned with axis A. Holes **78a** and **78b** (referred to either individually or collectively as hole(s) **78**) are defined transversely through sleeve **72** and intersect hole **76**. Holes **78** are disposed 180 degrees apart about axis A. Pins **70a** and **70b** are friction or interference fit into holes **78a** and **78b**, respectively.

[0051] Threaded shaft **80** includes an elongated body having a circular and hollow head **82** at its proximal end, and a non-circular threaded distal end **84**. Shaft **80** is also depicted in FIGS. **9A-9E**. The non-circular threaded distal end **84** includes opposing flat portions **85**. A circular blind hole **86** is formed in head **82** and is sized for receiving connector end **28** of driver shaft **20**, as is shown in FIG. **3C**. Holes **88a** and **88b** (referred to either individually or collectively as hole(s) **88**) are defined transversely through head **82** and intersect hole **86**. Holes **88** are disposed 180 degrees apart about axis A. Pins **70a** and **70b** are friction or interference fit into holes **88a** and **88b**, respectively, thereby captivating shaft **80** to sleeve **72**. Shaft **80** and sleeve **72** may be combined into a single unitary component. To that end, shaft **80** and sleeve **72** may be referred to herein more generally as a shaft.

[0052] In summary, and as best depicted in FIG. **3C**, pins **70a** and **70b** are positioned through holes **78a** and **78b** of sleeve **72** as well as holes **88a** and **88b** of shaft **80**, respectively, thereby non-rotatably connecting shaft **80** and sleeve **72**. Also, as noted above, sleeve **72** is non-rotatably connected to driver shaft **20**, and driver shaft **20** is non-rotatably connected to driver **14**. Accordingly, rotation of driver **14** results in rotation of threaded shaft **80** with respect to the stationary housing **60**. The radially outward portion of pins **70a** and **70b** are positioned and travel within ramps **66a** and **66b**, respectively, of stationary housing **60**.

[0053] It is conceivable that pins **70** could be mounted to the interior surface of housing **60**, and ramps **66a** and **66b** could be disposed on sleeve **72** (assuming that shaft **80** and sleeve **72** are fixed together).

[0054] Referring now to FIG. **2C**, door **12** may be a door for an HVAC unit, as noted above. HVAC unit doors, which are typically thick and filled with insulation, can have wide thickness tolerances. To accommodate for the thickness tolerance of the width D2 of door **12**, a longitudinal gap having a dimension D1 is provided between the distal tip **28a** of driver shaft **20** and the distal surface **86a** of blind hole **86** (i.e., in an assembled form of latch **10**). So long as driver shaft **20** remains rotationally keyed (i.e., non-rotatable) with sleeve **72**, dimension D1 can vary without affecting operation of latch **10**. For example, if door **12** were slightly wider than that which is shown, dimension D1 would be greater than that shown in FIG. **2C**, however, driver shaft **20** would be rotationally keyed with sleeve **72** and operate as intended. Conversely, if door **12** were slightly narrower than that which is shown, dimension D1 would be less, however, driver shaft **20** would be rotationally keyed with sleeve **72** and latch **10** operate as intended. Thus, latch **10** can be considered modular because it compensates for the thickness tolerance of the door **12**. Also, the length of driver shaft **20** may vary from that which is shown in order to accommodate door thicknesses outside of the above-described thickness tolerance.

[0055] Referring now to FIGS. 1C and 5, a pawl **90** is mounted to threaded distal end **84** of shaft **80** by two nuts **92**. Pawl **90** includes an L-shaped bracket **91**, and a roller cam **94** mounted to bracket **91** by a fastener **96**. As is known in the art, roller cam **94** is configured to interact with a frame (not shown) to which door **12** is movably attached. In the closed and locked state of latch **10** shown in FIGS. 3A-3C, roller cam **94** as well as door **12** is compressed against the frame. And, in the unlocked state of latch **10** shown in FIGS. 1A-1C, roller cam **94** is detached from the frame.

[0056] Pawl **90**, sleeve **72**, shaft **80**, pins **70** and roller cam **94** may together be considered as a pawl assembly or, more generally, as a pawl.

[0057] According to one method of operating latch **10**, starting from FIGS. 1A-1C, latch **10** is initially positioned in an unlocked state, for example. Lock plug **40** in in an unlocked state. In the unlocked state, driver **14** is rotated to a position where pawl **90** and its roller cam **94** do not register or interfere with the frame (not shown) such that door **12** can be moved to an open position with respect to the frame. In the unlocked state, pins **70** are positioned within distal portions **67b** of respective ramps **66**. When pins **70** are positioned within distal portions **67b** of respective ramps **66**, pawl **90** and its roller cam **94** are positioned far from the rear side **12b** of door **12** such that a gap or distance **C1** (FIG. 1C) is defined between the proximal side of roller cam **94** and the rear side **12b** of door **12**. Stated differently, at distance **C1**, pawl **90** is not compressed against the frame to which the door **12** is connected.

[0058] A user then moves latch **10** from the unlocked state to the partially unlocked/locked state shown in FIGS. 2A-2C. In the partially unlocked/locked state, driver **14** is rotated toward the locked state to a position where pawl **90** and its roller cam **94** register or interfere with the frame (not shown) such that door **12** cannot be moved to a fully open position. The door **12** can, however, be moved by distance **C1** toward the open position until the proximal side of roller cam **94** contacts the frame.

[0059] Upon rotating driver **14** from the unlocked state to the partially unlocked/locked state shown in FIGS. 2A-2C, driver shaft **20**, pins **70**, sleeve **72** and pawl **90** rotate simultaneously along with driver **14** (due to the above-described non-rotatable connections) with respect to the stationary housing **60**. As driver **14** is rotated, pins **70** slide within their respective ramps **66**. Specifically, pins **70** slide from distal portions **67b** to intermediate portions **67c**. At this stage, rotation of driver does not yet cause translation of pawl **90** along axis A and toward rear side **12b** of door **12**. Thus, in the partially unlocked/locked state shown in FIGS. 2A-2C, the distance **C1** is defined between the proximal side of roller cam **94** and the rear side **12b** of door **12**, and pawl **90** is not compressed against the frame to which the door **12** is connected.

[0060] The user continues to rotate driver **14** in the same rotational direction to the locked state shown in FIGS. 3A-3C. Specifically, as driver **14** is rotated further, pins **70** slide from intermediate portions **67c** to proximal portions **67a** of their respective ramps **66**. Moving pins **70** through proximal portions **67a** causes pawl **90** to simultaneously rotate and translate toward rear side **12b** of door **12**. Specifically, the geometry of proximal portions **67a** (which extend along the axis A in the proximal direction), as well as the pinned interface between housing **60**, sleeve **72** and shaft **80**, forces sleeve **72** and shaft **80** to move in a proximal direction toward rear side **12b** of door **12**. Once the pins **70** reach the terminal end of proximal portions **67a**, latch **10** is maintained in a fully-compressed and locked state. At this stage, roller cam **94** registers with the door frame to prevent door **12** from being opened. Also, the distance **C2** between the proximal side of roller cam **94** and the rear side **12b** of door **12** may be zero, or, more generally, less than the distance **C1**. Stated differently, at distance **C2**, pawl **90** is compressed against the frame. In a fully-compressed state of the latch **10**, any seal at the interface between door **12** and its frame (not shown) is also fully compressed.

[0061] If the user then rotates driver **14** in the opposite rotational direction (unlock direction) from the locked state shown in FIGS. 3A-3C to the partially unlocked/locked state shown in FIGS. 2A-2C, pins **70** move from proximal portions **67a** to intermediate portions **67c** of ramps **66**.

Consequently, pawl **90** rotates toward (but not too) the unlocked state and pawl **90** also translates in the distal direction (due to the geometry of proximal portions **67a**). At this stage, the distance **C1** is defined between the proximal side of roller cam **94** and the rear side **12b** of door **12**.

[0062] A detent or stop feature is defined at intermediate portions **67c** signaling to the user (via tactile feel) that door **12** is partially unlocked at this stage and the compartment to which door **12** is attached may be de-pressurizing. The user may stop rotating driver **14** at this point. More particularly, if the compartment to which door **12** is attached is pressurized, then moving driver **14** to the partially unlocked/locked state shown in FIGS. **2A-2C** will depressurize the compartment while preventing door **12** from suddenly moving to the fully open position as a result of the depressurization. Specifically, pawl **90** would contact the door frame of the compartment and prevent the door from fully opening while allowing door **12** to open by a limited distance (i.e., distance **C1**) for depressurization purposes. In the absence of intermediate portions **67c**, unlocking latch **10** could result in door **12** rapidly and unexpectedly opening due to depressurization of the compartment.

[0063] The user then continues to rotate driver **14** in the opposite rotational direction (i.e., the unlock direction) from the partially unlocked/locked state shown in FIGS. **2A-2C** to the unlocked state of FIGS. **1A-1C**. Continuing to rotate driver **14** in the opposite rotational direction causes pins **70** to move from intermediate portions **67c** to distal portions **67b** of ramps **66**. Consequently, pawl **90** rotates to the unlocked state shown in FIG. **1B**, but pawl **90** does not translate along axis **A** any further from its position shown in FIG. **2B**. Thus, the distance **C1** is defined between the proximal side of roller cam **94** and the rear side **12b** of door **12**. The latch **10** is again in an unlocked state.

[0064] FIGS. **12A** and **12B** depict a multi-point latch **100** for securing door **12**. Multi-point latch **100** comprises latch **10** (the same latch as described above) and two separate pawl sub-assemblies **33a** and **33b**. Pawl sub-assemblies **33a** and **33b** are connected to latch **10** by way of a linkage **102**. Linkage **102** interconnects latch **10** to pawl sub-assemblies **33a** and **33b** such that pawl sub-assemblies **33a** and **33b** move synchronously with pawl sub-assembly **33** of latch **10**. Accordingly, rotating latch **10** to the locked state causes pawl sub-assemblies **33a** and **33b** to also move to the locked state. And, rotating latch **10** to the unlocked state causes pawl sub-assemblies **33a** and **33b** to move to the unlocked state.

[0065] It should be understood that pawl sub-assemblies **33a** and **33b** are mounted to door **12** in the same manner as pawl sub-assembly **33** of latch **10**. Pawl sub-assemblies **33a** and **33b** are not directly connected to a driver sub-assembly **11**. Instead, pawl sub-assemblies **33a** and **33b** are indirectly connected to driver sub-assembly **11** of latch **10** via linkage **102**. It can be appreciated that pawl sub-assembly **33** may be used independently and without driver sub-assembly **11** directly mounted thereto.

[0066] Pawl sub-assemblies **33a** and **33b** are structurally and functionally equivalent to pawl sub-assembly **33** of latch **10**, thus, pawl sub-assemblies **33** are a modular feature of latch **10** that can be used together to perform a latching operation.

[0067] FIGS. **13A** and **13B** depict one mounting point on the top portion of linkage **102** of multi-point latch **100**. Linkage **102** comprises a rod **104** having a groove **106** formed therein and a series of holes defined through the thickness of rod **104** that intersect groove **106**. A hole **120** and an elongated slot **121** are formed through rod **104**. An L-shaped clip **108** is positioned at the top end of groove **106**. Clip **108** includes a first hole **110** for receiving a pin **112** extending from a cam **114a** and a second hole or slot **116** through which a fastener **118** passes through for mounting to a threaded hole **120** in rod **104**. Groove **106** and fastener **118** together prevent clip **108** from rotating or translating with respect to rod **104**.

[0068] Cam **114a** includes a pin **112**, as described above, as well as a non-circular hole **122** having two opposing flat portions **124**. An axis of rotation **B** of cam **114a** passes through hole **122**. Hole **122** is configured to receive non-circular threaded distal end **84** of shaft **80** in a tight form fitting manner. One or more fasteners **115**, such as hex nuts, are employed to fixedly connect cam **114a** to

threaded distal end **84**. By virtue of the non-circular connection between cam **114a** and shaft **80**, those components are non-rotatable with respect to each other (i.e., they rotate together). Pin **112** is positioned through slot **121** and hole **110** and is connected to a clip **126**. Clip **126** connects the free end of pin **112** to L-shaped clip **108** and prevents the free end of pin **112** from disengaging from clip **108**. Cam **114a** can rotate with respect to rod **104**.

[0069] Cam **114a** is mounted to shaft **80** that is associated with pawl sub-assembly **33a**. A second cam **114b** is mounted to the bottom end of rod **104** using a clip **108** in the same manner as cam **114a**. Cam **114b** is mounted to shaft **80** that is associated with pawl sub-assembly **33b**. A third cam **114c** is mounted to the longitudinal center of rod **104** in the same manner as cams **114a** and **114b**. Cam **114c** is mounted to shaft **80** that is associated with pawl sub-assembly **33** of latch **10**. Cams **114a-114c** are structurally and functionally equivalent and operate in the same manner.

[0070] Hole **120** and elongated slot **121** are associated with cam **114a**, and, although not shown, it should be understood that, another hole **120** and slot **121** are associated with cam **114b**, and yet another hole **120** and slot **121** are associated with cam **114c**. Elongated slots **121** and **116** together accommodate for variances in the vertical length and position of the various components of linkage **102** as well as the pawl sub-assemblies **33** to which linkage **102** is connected. While it is possible, it is not intended to have a vertical adjustment at cam **114c**, which is the driving location, because there exists a tight diametrical fit between the rod **104** and pin **112** at cam **114c**. Adjustment due to tolerance issues may only be required at the remote latches of pawl sub-assemblies **33a** and **33b**.

[0071] In operation, rotating driver **14** of latch **10** causes pawl **90** of pawl sub-assembly **33** to move between the locked and unlocked states, as was described in detail above. Shaft **80** moves with pawl **90**, as was described above. Rotation and translation of shaft **80** of latch **10** (as described above) also causes simultaneous rotation of cam **114c** about axis A as well as resultant translation of cam **114c** along axis A. Rotation and translation of cam **114c** causes simultaneous rotation of rod **104** about axis A as well as resultant translation of rod **104** along axis A. Rotation and translation of rod **104** causes simultaneous rotation of cams **114a** and **114b** about their respective axes B as well as resultant translation of cams **114a** and **114b** along their axes B. Rotation and translation of cams **114a** and **114b** causes simultaneous rotation of shafts **80** and pawls **90** of pawl subassemblies **33a** and **33b** about their respective axes B as well as resultant translation of shafts **80** and pawls **90** of pawl subassemblies **33a** and **33b** along respective axes B. Accordingly, in summary, rotation and translation of shaft **80** of latch **10** causes simultaneous rotation and translation of three separate pawls **90**, i.e., pawls **90** of pawl subassemblies **33**, **33a** and **33b**. Each pawl **90** provides a separate point of contact for latching door **12** in a closed and locked state.

[0072] A handle **130** is either connected to or extends from cam **114c**. Alternatively, handle **130** extends from pawl **90** of latch **10**. Handle **130** is non-rotatably connected to shaft **80**. Multi-point latch **100** can be operated using handle **130** in the same manner as described above with respect to driver **14**. In other words, rotation of handle **130** causes rotation and translation of three pawls **90** between locked and unlocked states. Handle **130** is provided on the rear side **12b** of door **12** to prevent a user from inadvertently being locked inside of a compartment, such as a freezer or refrigerated compartment.

[0073] While preferred embodiments of the invention have been shown and described herein, it will be understood that such embodiments are provided by way of example only. Numerous variations, changes and substitutions will occur to those skilled in the art without departing from the spirit of the invention. Accordingly, it is intended that the appended claims cover all such variations as fall within the spirit and scope of the invention.

Claims

1. A pawl assembly for locking a door, said pawl assembly comprising: a housing that is configured to be fixedly mounted to the door, and a pawl that is movably connected to the housing and is

configured to both rotate and translate with respect to the housing, wherein, in an unlocked position of the pawl assembly, the pawl is positioned to permit opening of the door, and, in a locked position of the driver, the pawl is positioned to prevent opening of the door, and wherein the pawl is configured to move closer to the door upon moving the pawl assembly from the unlocked position to the locked position.

2. The pawl assembly of claim 1, wherein the pawl is fixedly connected to a shaft of the pawl assembly.

3. The pawl assembly of claim 2, wherein the shaft of the pawl assembly includes a threaded shaft that is configured to be connected to the pawl, and a sleeve that is non-rotatably connected to the threaded shaft.

4. The pawl assembly of claim 2, wherein a pin associated with one of the shaft of the pawl assembly and the housing is engaged with a ramp disposed on the other of the shaft and the housing, such that motion of the pin along the ramp results in rotation and translation of the shaft and the pawl that is connected to the shaft.

5. The pawl assembly of claim 4, wherein the ramp is defined on the housing and the pin extends from the shaft.

6. The pawl assembly of claim 4, wherein the ramp includes a detent that is defined between two terminal ends of the ramp, wherein the detent defines an intermediate fixed position for the pin between the two terminal ends of the ramp, and wherein when the pin is positioned in the detent, the door is permitted to open by a limited distance.

7. The pawl assembly of claim 4, wherein the ramp extends in a direction along the housing that has a circumferential component as well as an axial component.

8. The pawl assembly of claim 1, wherein the pawl is configured to rotate about an axis and translate along the axis.

9. A driver that is configured to be rotatably mounted to a door, said driver comprising: a handle that is capable of being grasped by a user, an opening formed in the driver, a fastener positioned within the opening for mounting the driver to the door, a cover that is removably mounted to the driver and is positioned over the opening to conceal a head of the fastener, the cover defining a relief in the form of an undercut on a surface thereof, a lock plug being rotatably mounted to the driver between an unlocked state and a locked state, wherein in the unlocked state of the lock plug, the driver can be rotated with respect to the door, and in the locked state of the lock plug, the driver is prevented from rotating with respect to the door, and a projection extending from the lock plug, the projection being non-rotatable with respect to the lock plug, wherein in a locked state of the lock plug, the projection is positioned within the relief to prevent removal of the cover from the driver and consequent access to the fastener, and in an unlocked state of the lock plug, the projection is not positioned within the relief to permit removal of the cover from the driver and allow consequent access to the fastener.

10. The driver of claim 9, wherein the projection has a non-circular perimeter and extends about an entirety of the lock plug.

11. The driver of claim 9, wherein the relief is formed on a protrusion of the cover that is positionable within the opening formed in the driver.

12. The driver of claim 11, wherein the protrusion is axially aligned with the fastener.

13. The driver of claim 9, wherein the cover includes an opening through which the lock plug is mounted.

14. The driver of claim 9, wherein the lock plug is configured to rotate with respect to the cover and the driver.

15. The driver of claim 9, further comprising: a driver shaft that is non-rotatably connected to the driver such that said driver shaft rotates along with said driver.

16. A latch comprising the driver of claim 15 and a pawl assembly that is configured to be mounted to an opposite side of the door, said pawl assembly comprising (i) a housing that is configured to be

fixedly mounted to the door, and (ii) a pawl that is movably connected to the housing and is configured to both rotate and translate with respect to the housing in response to rotation of the driver shaft, wherein, in an unlocked position of the driver, the pawl is positioned to permit opening of the door, and, in a locked position of the driver, the pawl is positioned to prevent opening of the door, and wherein the latch is configured to translate the pawl closer toward the door upon rotating the driver from the unlocked position to the locked position.

17. A multi-point latch system comprising the latch of claim 16 and a plurality of said pawl assemblies.
